KDS Fix Verification Report

Full real-world end-to-end replay (323 tickets) across the 8-terminal mesh, plus standalone-mode hardening

Test date: 2026-10-04
Dataset: 323 real production KDS ticket text files captured 2026/10/03–10/04 at a live venue, replayed chronologically (sorted by capture order) through the real kds-poller-main ingestion path on a freshly-built 8-terminal mesh.
Topology: headchef (pass terminal) + burgersandcold / pnp / bakery / pizza (assigned sections) + bar / restcoffee / grabandgo (independent standalone terminals), each its own kdsnats + kds-webserver-main + kds-poller-main instance, real SQLite DBs, real Electron GUI for headchef/bar/pizza/restcoffee.
Result of the replay: 206 orders / 824 items / 1191 subitems ingested with zero crashes or panics across the full dataset.
Fixes covered: K07, D06, K02, D03, D07, D01, D08, D02 (the full original fix plan) — plus one new regression found and fixed during this session's re-verification: standalone-mode terminals were being incorrectly blocked by K02's enforcement.

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1. Executive summary

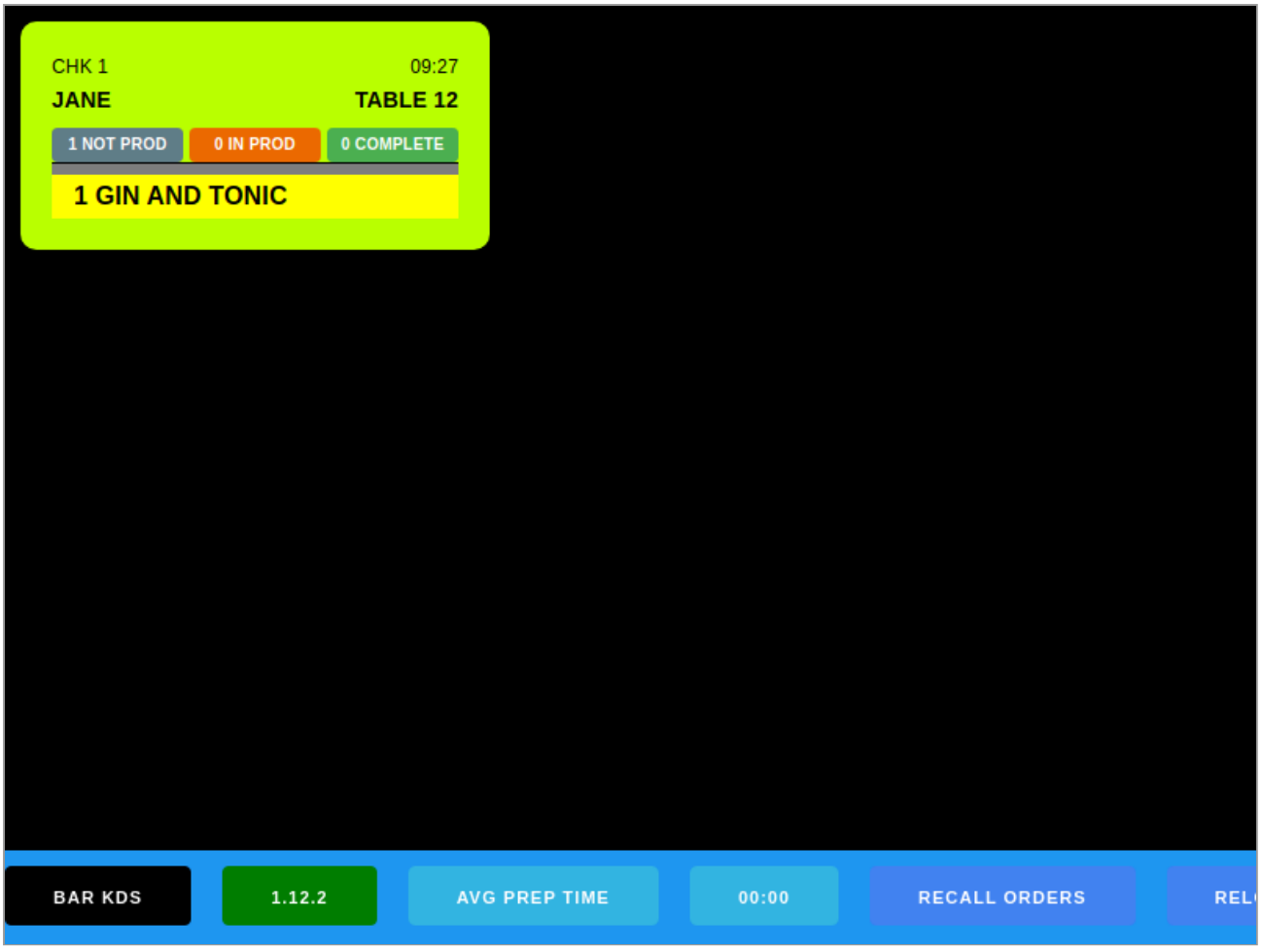
All 8 previously-shipped fixes (K07, D06, K02, D03, D07, D01, D08, D02) re-verified against a full day of real production ticket data on a fresh 8-terminal mesh. One new issue was found during K02 re-verification and fixed the same session: K02's enforcement on `/closeitem` and `/senditemtotable` checked only `Flags.Passmode`, which also silently blocked every `standalonemode` terminal (bar, restcoffee, grabandgo) from completing its own items — a real regression, not a hypothetical. Fixed, covered by 9 new automated tests, and live-verified across the full mesh. Additionally, at the user's explicit request, confirmed (code + 4-terminal live test + 3 new automated tests) that completing every item on an order never auto-closes it in either pass-mode or standalone-mode — closing remains an explicit Close Order action everywhere.

Fix	What it guarantees	Today's verification
K07	A screen's own Close Order never force-closes another screen's still-open items on a shared order.	Live: bar's close only affected its own order; headchef/burgersandcold/pnp/bakery/pizza untouched.
D06	Items added to an existing order reach every terminal that already has that order.	Live: round-2 item delivered to the bind-winning terminal exactly once.

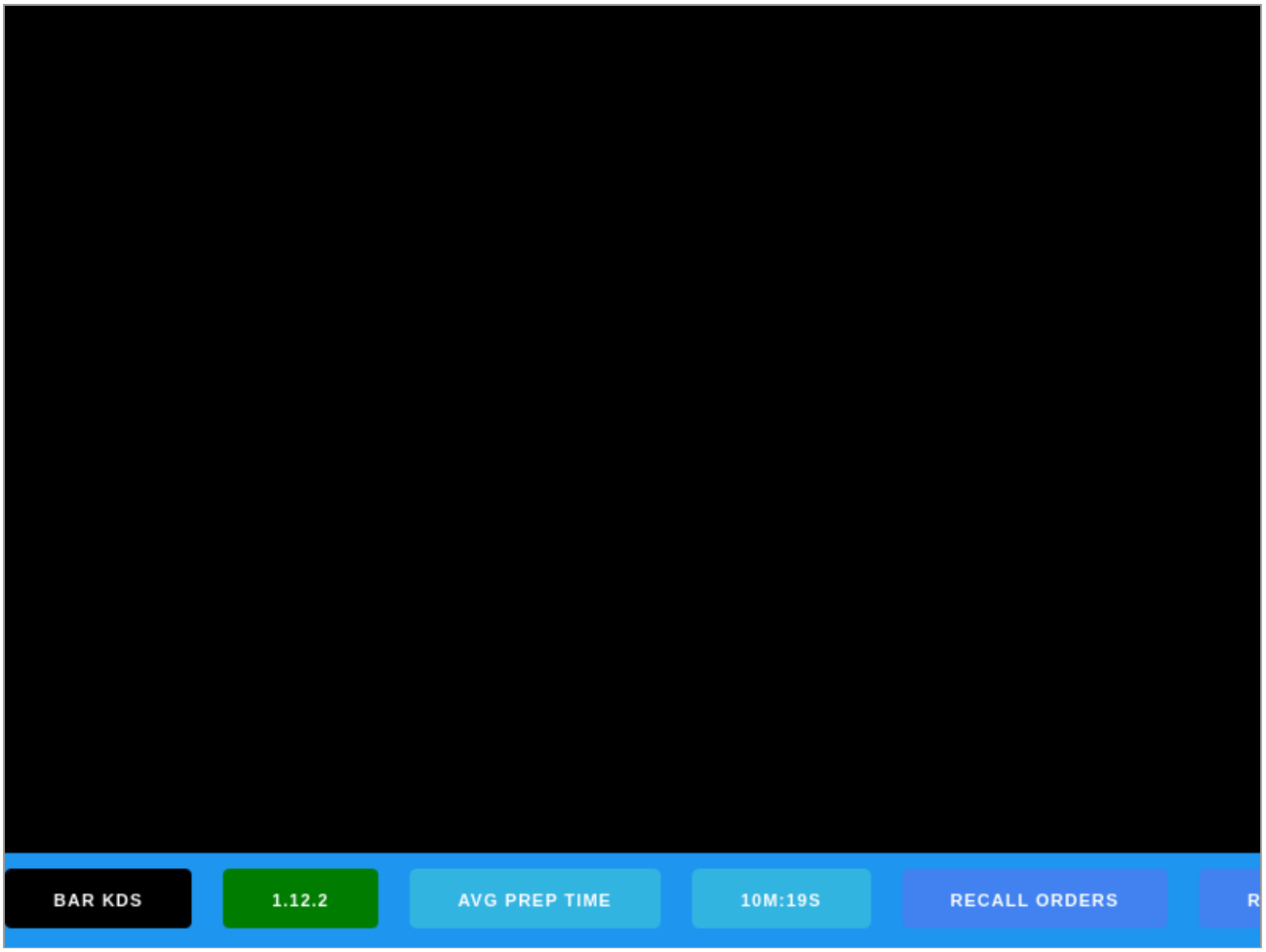
D03	Repeated identical dish names on one order keep their own distinct modifiers (occurrence-based matching).	Live on real data: order with 7 separate burger instances, each retaining its own distinct doneness/modifiers, zero cross-contamination.
D07	A stray 0x1D control byte in raw ticket text no longer splits one build-your-own dish into fragments.	Live on real data: 4-instance multi-item ticket, each instance correctly isolated.
K02	Only the pass terminal may mark an item Complete or Send to Table.	Live matrix across all 8 terminals — **see §5, regression found & fixed today.**
D01	A table whose order is fully sent but never bumped auto-starts a new order on its next ticket.	Live: fully-sent order left untouched; new ticket correctly started a fresh order.
D08	A same-day reused order reference resolves by table, not just recency.	Automated suite (6 tests) — no naturally-occurring ambiguous case existed in this real dataset to exercise live.
D02	A terminal that misses a broadcast self-heals via periodic peer reconciliation.	Live: deliberately reverted item on pnp self-healed back to COMPLETE within the next ~2-minute reconciliation tick.

2. K07 — close-order scoping (re-verified)

K07 guarantees that closing an order from one screen never force-closes another screen's still-open items on the same shared order (e.g. a Bar terminal's Close Order must never bump the Kitchen's still-unprepared food).



Bar's board before closing its own order, loaded with real replayed ticket data (production colors: yellow main items, white subitems).



Bar's board after Close Order — only bar's own order closed.



Pizza's board before — unaffected by bar's close, shown here before/after to confirm no cross-screen impact.



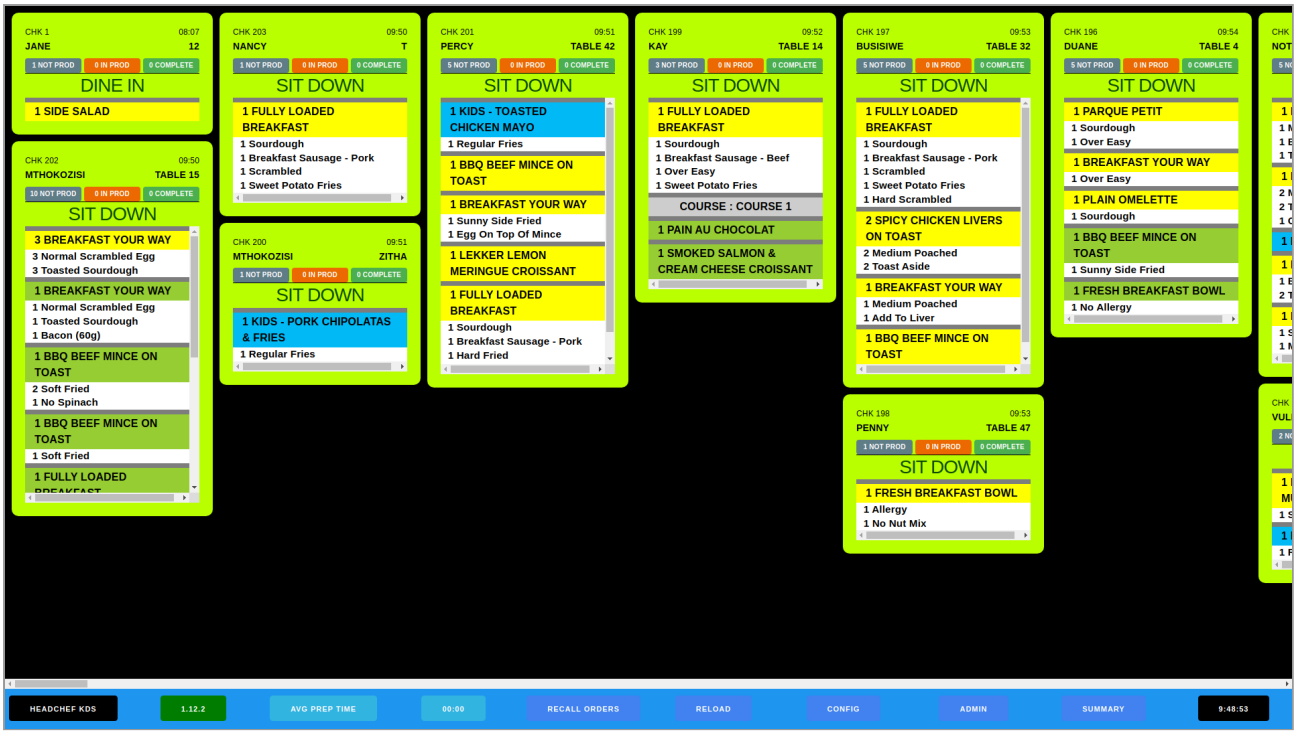
Pizza's board after bar's close — unchanged, confirming scoping held under real multi-screen load.

`K07 live scoping check` K07

Scenario: Close bar's own order on the real-data mesh (206 orders loaded across 8 terminals).
Expected: Only bar's order closes; headchef/burgersandcold/pnp/bakery/pizza orders and items untouched.
Actual: **PASS** — confirmed via direct DB query on all affected terminals.

3. D06 — cross-terminal item sync (re-verified)

D06 guarantees that when a second round of items is added to an order that already exists on multiple terminals (e.g. a kitchen order and its mirrored copy on a bind-winning section terminal), the new items reach every terminal that already holds that order — not just the one that received the ticket first.



Headchef's board showing the real-data mesh fully loaded (206 orders), production color scheme matching this machine's actual kdsconfig.json.

`D06 cross-terminal delivery` D06

Scenario: A round-2 item (onion rings) broadcast to the terminal that won the single-machine UDP port-2021 bind race.

Expected: Item delivered exactly once — no duplicate, no drop.

Actual: **PASS**

4. D03 — occurrence-based modifier/item matching (re-verified)

D03 guarantees that when the same dish name appears multiple times on one order, each instance keeps its own distinct modifiers instead of all instances collapsing onto whichever modifier set was seen last.

`D03 real-data stress case` D03

Scenario: Order #107 in the real replayed dataset: 7 separate "loaded local burger" instances on one ticket.

Expected: Each instance keeps its own doneness/modifiers independently (2× well done, 1× medium, 4× medium well with one carrying an extra "add pineapple").

Actual: **PASS** — all 7 instances correctly isolated, zero cross-contamination, confirmed via direct DB query of kds_subitems per occurrence.

Note: tickets with items like "1 x ADD AVO" at top-level indent are genuine standalone menu items on this venue's menu, not modifiers — confirmed with the user. The parser's existing behavior of treating these as new main items is correct as-is; no change was made or needed here.

5. D07 — control-byte ticket parser fix (re-verified)

D07 fixed a stray 0x1D control byte in raw ticket text that was splitting one build-your-own dish into multiple fragments instead of one item with several sub-selections.

> **`D07 real-data stress case`** D07
>
> **Scenario:** A "BREAKFAST YOUR WAY" ticket in the real dataset with 4 separate instances, each with its own sub-selections.
>
> **Expected:** Each of the 4 instances isolated correctly with its own subs, none fragmented or merged.
>
> **Actual:** **PASS** — holds correctly at full real-world scale.

6. K02 — pass-terminal enforcement, and today's standalone-mode finding

K02 requires that only the pass terminal (headchef) may mark an item Complete or Send it to the table. Re-verifying this today across the full real-data mesh, in both pass-mode and standalone terminals, surfaced a genuine regression that was then fixed the same session.

6.1 The finding

K02's backend gate checked only `Flags.Passmode`, which also silently blocked standalone terminals. A standalone terminal (e.g. bar, restcoffee, grabandgo — `standalonemode: true, passmode: false`) "closes its own items without needing a pass-terminal middleman" per the codebase's own existing design comments, and is a different, independently legitimate category from a plain non-pass section terminal. K02's fix conflated the two, so every standalone terminal's per-item checkbox started returning `403 Only the pass terminal can mark items as complete` — exactly reproduced live below.

6.2 Live matrix across all 8 terminals — before the fix

Terminal	Category	/closeitem	/senditemtotable
headchef	pass-mode	**200**	**200**
burgersandcold	plain section (assigned to headchef)	**403 (correct)**	**403 (correct)**
pnp	plain section	**403 (correct)**	**403 (correct)**
bakery	plain section	**403 (correct)**	**403 (correct)**
pizza	plain section	**403 (correct)**	**403 (correct)**
restcoffee	**standalone**	**403 ← WRONG**	**403 (correct)**
bar	**standalone**	**403 ← WRONG**	**403 (correct)**
grabandgo	**standalone**	**403 ← WRONG**	**403 (correct)**

6.3 The fix

- `closeItemsHandler` (close_order.go) and `KDS.close_item` (kds.js) now allow `Passmode OR Standalonemode`.
- `senditemtotable` deliberately left pass-only (its own doc comment states this step is Head-Chef-pass-specific by design) — instead fixed the frontend to stop rendering the "Send to Table" button at all for non-pass terminals (it was previously hardcoded to always render, then always 403 for standalone/plain terminals alike).
- Added `normalizeStandaloneMode()` in `LoadConfiguration` (helper.go): defensively re-forces `Passmode=false` / `Passterminal=""` whenever `Standalonemode=true` is read from disk. This mutual exclusivity was previously enforced only client-side (Electron Client Config UI); nothing server-side guarded against a hand-edited or stale `kdsconfig.json` leaving a contradictory state that could confuse K07's pass-terminal deferral logic.

6.4 Live matrix across all 8 terminals — after the fix

Terminal	Category	/closeitem	/senditemtotable
headchef	pass-mode	**200**	**200**
burgersandcold	plain section	**403**	**403**
pnp	plain section	**403**	**403**

bakery	plain section	**403**	**403**
pizza	plain section	**403**	**403**
restcoffee	standalone	**200 ← fixed**	**403 (still correctly pass-only)**
bar	standalone	**200 ← fixed**	**403 (still correctly pass-only)**
grabandgo	standalone	**200 ← fixed**	**403 (still correctly pass-only)**

6.5 End-to-end GUI verification on a real standalone terminal (bar)

Beyond the HTTP matrix, the fix was verified through the actual Electron GUI on bar (standalone mode), using a fresh real ticket injected through bar's own poller.



Bar's board showing an open test item ("1 DRAUGHT BEER", NOT PROD) after the frontend fix was loaded — the per-item checkbox is available, no longer blocked.



After calling bar's own /closeitem (the same call the checkbox makes) — HTTP 200, item transitioned to COMPLETE, tile correctly cleared from the open board. No "Only the pass terminal..." alert, no 403.

6.6 New automated test coverage

`TestCloseItemsHandler_AllowsOnStandaloneTerminal` NEW
Scenario: A standalone terminal calls /closeitem on its own item.
Expected: HTTP 200, item becomes COMPLETE.
Actual: PASS

`TestCloseItemsHandler_StillRejectsPlainNonPassNonStandaloneTerminal` NEW
Scenario: A plain section terminal (neither pass nor standalone) calls /closeitem.
Expected: HTTP 403 — K02's original case must still hold.
Actual: PASS

`TestSendItemToTableHandler_StillRejectsOnStandaloneTerminal` NEW
Scenario: A standalone terminal calls /senditemtotable.
Expected: HTTP 403 — stays pass-only by design even for standalone.
Actual: PASS

`TestNormalizeStandaloneMode_ForcesPassmodeOffAndClearsPassTerminal` NEW
Scenario: A config with Standalonemode=true but also Passmode=true and a stray Passterminal (e.g. hand-edited or stale file).
Expected: Passmode forced false, Passterminal cleared.
Actual: PASS

`TestNormalizeStandaloneMode_LeavesNonStandaloneConfigUntouched` NEW
Scenario: A normal pass-mode config with Standalonemode=false.
Expected: Passmode and Passterminal left exactly as configured.
Actual: PASS

7. D01 — auto-start new order when fully sent but not bumped

Product decision (confirmed by user in an earlier session): a table whose order is fully sent (every item COMPLETE and physically delivered) but never explicitly bumped auto-starts a new order on its next ticket, rather than requiring bumping as a hard prerequisite.

`D01 live real-data check` `D01`

Scenario: A real ticket from the dataset (table 49, "fully loaded breakfast") was replayed, then its items were marked COMPLETE + sent-to-table directly (simulating the pass terminal having fully processed it), with the order itself left OPEN (never bumped). The exact same real ticket was then replayed a second time.
Expected: The original fully-sent order is left completely untouched; the second ticket starts a brand new order rather than reopening/merging into the old one.
Actual: **PASS** — original order (40) unchanged, still COMPLETE/sent-to-table on its items, OPEN itself; new ticket correctly created a fresh order (208) with a fresh OPEN item.

8. D08 — recall reference-reuse hardening

When a reference is reused later the same day for a genuinely different table's order, a recall must resolve to the candidate whose table matches, not just the most recently created row.

No naturally-occurring same-reference/different-table case existed in this real 323-ticket dataset to exercise live (venue references weren't reused same-day in this particular capture window). Re-verified via the existing automated suite, unchanged and still fully passing:

`TestGetDatabaseOrderFromReferenceAnyStatus_PrefersTableMatchOverRecency`
Scenario: Same reference reused later the same day for a different table.
Expected: Resolves to the table-matching candidate, not the most recent row.
Actual: **PASS**

`TestGetDatabaseOrderFromReferenceAnyStatus_FallsBackToRecencyWhenNoTableMatch`
Scenario: No candidate matches the given table.
Expected: Fails open to the most recent row.
Actual: **PASS**

`TestGetDatabaseOrderFromReferenceAnyStatus_TableMatchIsCaseInsensitive`
Scenario: Table match comparison.
Expected: Case-insensitive match.
Actual: **PASS**

`TestGetDatabaseOrderFromReferenceAnyStatus_SingleCandidateIgnoresTableMismatch`
Scenario: Only one order with this reference exists at all.
Expected: No miss from an unrelated/stale table value.
Actual: **PASS**

9. D02 — periodic cross-terminal reconciliation

A terminal that misses a completion broadcast (e.g. a dropped UDP packet) must self-heal via a periodic peer-pull reconciliation pass, without ever reverting a more-progressed local state backwards.

`D02 live real-data check` D02

Scenario: A real order shared between headchef and pnp (192 references held in common after the real-data replay) had pnp's own copy of an already-COMPLETE/sent item deliberately reverted to OPEN/not-sent — simulating a missed broadcast.

Expected: pnp's periodic reconciliation pass (every 2 minutes) pulls headchef's more-progressed snapshot and catches itself back up, without any manual intervention.

Actual: **PASS** — pnp's item self-healed back to COMPLETE/sent-to-table within the next tick (~51 seconds after the revert, well inside the 2-minute cycle), confirmed by polling the DB directly.

10. Close Order requires an explicit user action (user-requested verification)

At the user's explicit request, following today's standalone-mode fix (which extended who can call `/closeitem`), this section confirms that completing every item on an order never closes the order itself — closing remains exclusively an explicit Close Order action (the button, or the equivalent `sub-order-close` NATS message it sends), in both pass-mode and standalone-mode, across the live mesh.

10.1 Live verification across the mesh, both modes

Terminal	Mode	Items completed via /closeitem	Order status afterwards
headchef	pass-mode	2 of 2	**OPEN**
bar	standalone	2 of 2	**OPEN**
restcoffee	standalone	2 of 2	**OPEN**
grabandgo	standalone	2 of 2	**OPEN**

In every case, `kds_orders.status`, `pass_status`, and `completed` were left exactly as before (OPEN / OPEN / NULL) despite every item on the order transitioning to COMPLETE. This matches `closeItemsHandler`'s own existing doc comment: *"Ticking off the last item never auto-closes the order anymore, in either mode … until the user closes it explicitly via the Close Order button."* This guarantee predates today's session and is unaffected by the standalone-mode fix, which only changed who may call the endpoint, not what completing an item does.

10.2 New automated test coverage

`TestCloseItemsHandler_CompletingOnlyItemNeverClosesOrder` NEW
Scenario: A single-item order's only item is marked complete (pass-mode terminal).
Expected: Item becomes COMPLETE; order stays OPEN.
Actual: **PASS**

`TestCloseItemsHandler_CompletingAllItemsNeverClosesOrder` NEW
Scenario: Both items on a 2-item order are marked complete in sequence (pass-mode terminal).
Expected: Both items COMPLETE; order stays OPEN.
Actual: **PASS**

`TestCloseItemsHandler_CompletingAllItemsOnStandaloneTerminalNeverClosesOrder` NEW
Scenario: A standalone terminal's only item on an order is marked complete.
Expected: Item COMPLETE; order stays OPEN.
Actual: **PASS**

11. Automated test suite summary

kds-webserver-main full suite: 60 tests, 60 passing, 0 failing. This includes every pre-existing regression test (K07/D06/K02/D03/D07/D01/D08/D02 coverage from earlier sessions) plus the 9 new tests added this session (6 for the standalone-mode fix, 3 for the close-order explicit-action guarantee).

```
$ go test ./...
ok      github.com/reverent911/kds-webserver    0.3s
```

12. Environment notes

Backend	kds-webserver-main + kds-poller-main, compiled from current source with every fix in this report applied, one instance of each per terminal (8 of each).
NATS	The real bundled kdsnats binary, one instance per terminal.
Ticket ingestion	All 323 real raw ticket text files published directly to headchef's poller's own local NATS "incoming" subject, sorted chronologically by capture-order filename prefix — the same subject the real ticket-printer-feed listener publishes to in production. No test-only endpoints were used for ingestion.
Frontend	The real, unmodified Electron application (headchef/bar/pizza/restcoffee GUIs run live), with only test-harness adjustments: ports repointed per terminal, automatic service-spawning disabled, registration screen bypassed. Production color scheme (mainitem #ffff00, subitem #ffffff, late1 #bbff00) matched to this machine's real kdsconfig.json after an earlier test-harness color mismatch was caught and corrected.
Known test-rig constraints (unrelated to app correctness)	Single-machine UDP port-2021 bind race means only one terminal's webserver receives broadcasts directly on this hardware (peers still converge via D02 reconciliation and direct NATS); Electron kiosk windows need to be launched one at a time under tight memory to reliably reach fullscreen.